

Advanced Human-Computer Interaction: Architectural Paradigms, Empirical Methodologies, and Scientific Publishing Protocols

The Evolution of User Interface Architecture

The landscape of user interface design is undergoing a fundamental transition from static, screen-centric layouts to dynamic, intent-driven, and multimodal environments.¹ Historically, interfaces required human adaptation to rigid system constraints.¹ Contemporary paradigms, however, utilize adaptive systems that actively collaborate with human users, adjusting both form and function to fit real-time user context, psychological states, and implicit behavioral signals.¹

Emerging Paradigms of Intent-Based and Generative User Interfaces

Historically, user interfaces were constructed as static screen sequences, fixed pathways, and customer funnels.¹ Modern interactive systems are replacing these predefined journeys with generative experiences designed around outcomes and user intent.¹ Generative user interfaces do not rely on pre-built templates or static page layouts.¹ Instead, they utilize real-time personalization, predictions, and behavioral signals to decide what to show, emphasize, and adapt.¹ Menus and product lists reorder dynamically based on habits, and system alerts automatically adjust to current user activity levels.² Under this model, when a user states a specific goal, the interface dynamically builds the necessary screen on demand to facilitate task completion.²

To optimize these generative models, developers must categorize and capture user intent across multiple dimensions, which can be expressed explicitly through prompts and selections, or implicitly inferred from behavior.¹

Intent Type	Operational Objective	Interface Adaptation Strategy
Informational	Knowledge and information retrieval	Dynamic prioritization of content, text synthesis, and semantic search interfaces. ¹

Navigational	Directing to a specific digital or physical destination	Minimization of cognitive pathways, direct deep-linking, and adaptive search routing. ¹
Commercial	Evaluating alternatives within a purchasing lifecycle	Real-time comparative matrices, dynamic product filter reordering, and predictive recommendations. ¹
Transactional	Completing a specific action or purchase	Frictionless micro-interactions, context-aware payment sheets, and structural field reduction. ¹

This generative model shifts the role of the designer from managing spatial layouts and visual aesthetics to defining the underlying system logic, behavioral parameters, and flows of understanding.¹ While generative systems enhance interaction speed and personalization, they also introduce distinct risks regarding algorithmic bias, IP integrity, and user trust.² AI models trained on large datasets can propagate biased patterns or copy copyrighted styles, necessitating rigorous human-in-the-loop review processes.² Furthermore, unexplained interface changes can disorient users, which makes explainability features essential.² Providing simple text explanations (e.g., displaying a message like "Analyzing your data") helps mitigate user confusion and maintains system trust.² Additionally, because dynamic personalization relies heavily on user data, designers must navigate significant security and privacy adoption barriers, ensuring that users retain granular control over what data is collected and how it is used.²

Multimodal Interactions and Contextual Interfaces

As physical screens recede from being the exclusive focal point of computing, multimodal experiences are emerging to facilitate natural collaboration between humans and machines.¹ These systems synthesize voice (speech, tone, and intent), vision (gaze, gesture, and camera inputs), touch (force and haptic feedback), and contextual sensors (location, environment, and movement).¹ Emotionally aware interfaces reduce cognitive fatigue, build long-term trust, and treat components such as color, motion, copy, sound, and pacing as emotional instruments.¹ Designing these multimodal systems requires adherence to several core principles:

- **Asymmetric Modality Allocation:** Systems must avoid overwhelming the user by enabling all inputs simultaneously. Instead, the architecture should dynamically select and emphasize the single most appropriate mode for the immediate task and environment.¹
- **Frictionless Mode Switching:** Transitioning from one mode to another—such as

initiating a query with voice, glancing at a display, and confirming with a touch gesture—must occur without state resets, data loss, or user-perceived latency.¹

- **Graceful Degradation and Fallbacks:** Interaction modes frequently fail due to environmental constraints (e.g., voice input in high-decibel environments, gesture tracking in low-light conditions). The system must automatically deploy alternative modalities without disrupting the user flow.¹
- **Non-Visual Feedback Mechanisms:** In the absence of screen real estate, feedback must rely on precise audible, haptic, or temporal cues to keep the user aligned with the system status.¹

Human-AI Collaboration and Prompt Design

The transition toward collaborative human-machine systems shifts the interaction paradigm from direct control to mutual cooperation.¹ Within this collaborative framework, interfaces must assist users in formulating high-quality prompts and verifying generative outputs.¹ Research indicates that prompt engineering is a critical point of friction, which can be optimized using a structured checklist during interface design.¹

- **Empathy Framework:** The interface must guide the user to clarify who the target audience is, what prior knowledge they possess, and what emotional state they are likely experiencing.¹
- **Strategic Intent:** The system must assist in identifying the core problem, deciding if text or a visual representation is the most effective format, and defining the specific outcome desired.¹
- **Guardianship of Consequence:** The interface must prompt users to define boundaries, legal or ethical constraints, and verify sources that must be cited.¹
- **Verification Protocols:** The system must provide explicit, accessible mechanisms for the user to verify the generated output against real-world expectations and user needs prior to implementation.¹

Machine Experience Design

The proliferation of generative search engines and autonomous AI agents has introduced a new dimension to user interface engineering: Machine Experience (MX) design.¹ Digital platforms are no longer read exclusively by human eyes; they are parsed, cataloged, and summarized by LLM-driven agents long before a human ever arrives.¹ If a website presents messy structural signals or non-semantic code, these agents misinterpret the content, rendering the platform functionally invisible in generative search results.¹ Optimizing for machine readability requires embedding structural logic directly into the code and design systems.¹ This is achieved by:

- **Semantic Tokenization:** Implementing design tokens that describe functional roles (e.g., button-primary-background) rather than raw stylistic values (e.g., blue-500), enabling automated systems to infer the interactive weight of elements.¹
- **Component Metadata Documentation:** Embedding explicit descriptions within design libraries and codebases to detail *why* a component exists and *when* it should be utilized.¹

- **Explicit Relationship Mapping:** Utilizing semantic HTML and ARIA attributes to structurally link interactive elements (such as pairing form input fields with their exact programmatic labels) to prevent parsing errors by machine crawlers.¹
- **Structured Information Hierarchies:** Adhering to strict, predictable heading structures and standard UI patterns to ensure web spiders can rapidly parse and accurately represent the system's content.¹

Leading Research Forums and Emerging Academic Frontiers

Disseminating research in Human-Computer Interaction requires aligning scholarly contributions with appropriate academic forums and emerging thematic directions.³

Premier Conferences and Academic Forums

The peer-reviewed landscape of HCI consists of highly competitive conferences that serve as the primary venues for archival publications, tool demonstrations, and community-driven learning.³

Venue Name	Primary Audience & Focus	Core Key Themes & Paradigms	Typical Presentation Format
ACM CHI	Global academics, industry researchers, and HCI practitioners. ⁴	Broad, diverse, and interdisciplinary human-computer interaction topics. ³	Highly rigorous peer-reviewed papers, workshops, and tutorials. ⁴
ACM IUI	Computer scientists and AI researchers at the intersection of AI and HCI. ⁴	Intelligent user interfaces, human-AI co-creation, metaverse systems, and resilience. ⁴	Main track presentations, targeted workshops, and tool demonstrations. ⁴
ConveyUX	UX designers, product managers, and design strategists. ⁴	Industry adaptations to AI-driven workflows, accessibility, and strategy. ⁴	Hands-on workshops, case studies, and professional networking. ⁴

HUCAPP	European and international HCI researchers and software engineers. ⁶	Theoretical and practical foundations of human-computer interaction. ⁶	Academic papers, keynotes, and technical sessions. ⁶
UX360 Virtual Summit	Global user experience researchers and UX practitioners. ⁵	Quantitative and qualitative research planning, execution, and impact. ⁵	Case studies, interactive panels, and virtual round-tables. ⁵
Prompt:UX	AI engineers, product designers, and AI-UX practitioners. ⁵	AI integration into UX, workflow optimization, and generative UI strategies. ⁵	Keynotes, practical case studies, and networking receptions. ⁵
DDX Tokyo	Digital innovators, customer experience (CX) researchers, and UI designers. ⁵	Converging AI, strategic design, and customer insights for business impact. ⁵	Single-day panels, keynotes, and collaborative round-tables. ⁵
UXDX EMEA	Cross-functional product, design, research, and engineering teams. ⁷	Shifting from projects to product teams, features vs. business value in the AI era. ⁷	Case studies from leading brands, collaborative team workshops. ⁷
UX London	UX practitioners seeking equal parts inspiration and hands-on training. ⁷	Thematic focus: Discovery Day (research), Design Day (UI), and Delivery Day (systems). ⁷	Morning theoretical talks paired with afternoon hands-on workshops. ⁷
UX Scotland	Community-rooted UX and service designers in the civic and public sectors. ⁷	"Constraint as a Catalyst"—navigating legacy systems, budgets, and politics. ⁷	CPC-accredited workshops, discussions, and peer learning. ⁷

Emerging Research Streams and Workshop Frontiers

A deep analysis of the accepted workshops at recent premier conferences, such as ACM CHI, reveals the active research frontiers and conceptual debates within the scholarly community.¹⁰ These workshops serve as pre-archival incubation spaces where novel methodologies are evaluated and formalized into research agendas.⁸

- **The Sociomateriality of Empathy (EmpathiCH):** Over several iterations, researchers have sought to unpack how empathy is conceptualized, measured, and applied in HCI, transitioning from purely psychological views to a sociomaterial framework.¹⁰ This perspective examines how technology, social practices, and environmental contexts shape design practices, aiming to establish a sociomaterial taxonomy for empathy-centered design.¹⁰
- **Agentic Automation Experiences:** The rise of agentic AI systems—which coordinate multiple specialized agents behind a unified interface—presents challenges for human control and system transparency.¹⁰ Current research focuses on multi-agent transparency (understanding how choices are made across distributed agents) and preserving human agency during proactive automation.¹⁰
- **Generative UI and Adaptive Workspaces:** Scholars are actively investigating the systemic implications of Generative UI on HCI practices.¹⁰ This includes evaluating how designers navigate latent design spaces, build collaborative playgrounds, and transition from explicit pixel-level control to orchestrating dynamic layout logic.¹
- **Equitable and Intercultural AI:** The intersection of AI and cultural design systems is a growing research priority.¹⁰ Workshops focus on co-designing equitable, culturally grounded futures to ensure that generative models do not default to Western-centric interaction patterns or perpetuate historic biases.²
- **Trust Calibration and Cognitive Biases:** As AI systems evolve, human trust must be dynamically calibrated to prevent both over-reliance and under-reliance.¹⁰ Researchers study how interface feedback can leverage or mitigate cognitive biases to establish appropriate levels of human trust during high-stakes tasks.¹⁰

Mathematical Models and Cognitive Frameworks

HCI research relies on formal mathematical models and established psychological laws to predict user performance, evaluate layout efficiency, and explain cognitive behavior.¹²

Psychological Frameworks and Human Performance Laws

A comprehensive understanding of cognitive and behavioral laws is essential for predicting human performance during interface interaction.¹²

Psychological Law / Effect	Core Theoretical	Visual Design
----------------------------	------------------	---------------

	Proposition	Recommendation
Fitts's Law	Movement time to a target is a function of target distance and size. ¹²	Make critical action buttons larger and position them closer to natural cursor paths. ¹³
Hick-Hyman Law	Decision-making time increases logarithmically with choice count and complexity. ¹²	Minimize options during critical moments; utilize progressive onboarding to manage load. ¹⁴
Tesler's Law	Every system has an inherent level of complexity that cannot be reduced further. ¹²	Balance simplicity with function; avoid oversimplifying to the point of abstract confusion. ¹²
Jakob's Law	Users spend most of their time on other sites, expecting yours to function similarly. ¹¹	Adhere to established external design conventions, layouts, and standard UI patterns. ¹⁶
Von Restorff Effect	Visually distinct elements are more likely to be remembered by the user. ¹²	Utilize high contrast, distinct styling, or size variations to emphasize primary actions. ¹²
Miller's Law	Working memory can typically hold only seven (plus or minus two) distinct items. ¹²	Group content into logical, scannable chunks to ease information retention. ¹²
Aesthetic-Usability Effect	Visually pleasing interfaces are perceived as easier to use and more tolerant of errors. ¹²	Invest in aesthetic appeal to reduce user frustration during complex tasks. ¹²
Doherty Threshold	Computer response times under 400ms optimize user focus and decision speeds. ¹²	Optimize technical performance; use skeletons or spinners to manage perceived load. ¹²

Information-Theoretic Modeling and the Divergence of Fitts' and

Hick's Laws

Both Fitts's Law and the Hick-Hyman Law emerged from Shannon and Weaver's (1949) Information Theory, modeling human motor and cognitive behaviors by analogy to information transmission channels.¹⁵ Within this information-theoretic framework, human movements and choices are represented in bits of information.¹⁵

Paul Fitts described a method to mathematically calculate the movement time (T_M) required to acquire a target based on its distance (D) and width (W)¹³:

$$T_M = a + b \log_2 \left(\frac{2D}{W} + 1 \right)$$

where a and b are empirical constants determined via regression analysis.¹⁵ This model has achieved widespread adoption in HCI, serving as a cornerstone of Keystroke Level Modeling (KLM) and predicting physical target selection times across varying input devices and screen sizes.¹³

Conversely, the Hick-Hyman Law models decision-making time (T_R) as a logarithmic function of the number of options (n) presented to the user¹⁴:

$$T_R = b \log_2(n + 1)$$

where choice selection speed scales with the entropy of the decision stimulus.¹⁵ However, historical analysis reveals that the Hick-Hyman Law failed to gain significant momentum in empirical HCI research compared to Fitts's Law.¹⁵ This divergence is driven by several systemic factors¹⁵:

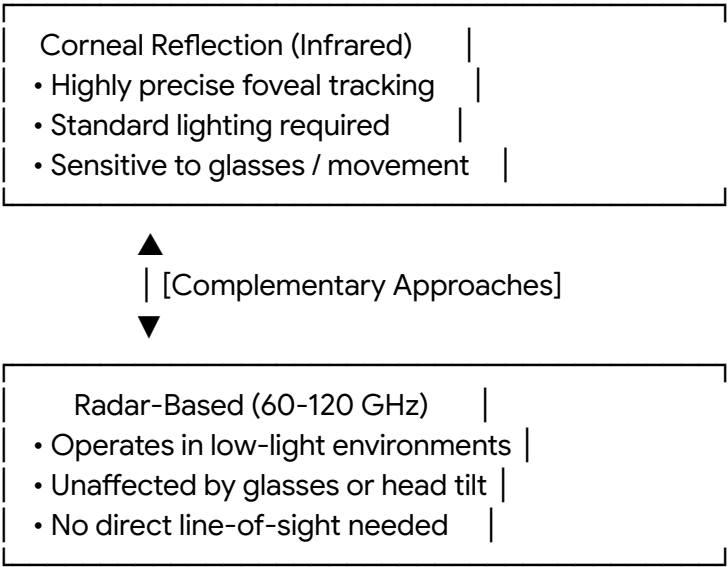
- **Visual Grouping and Categorization:** Modern user interfaces rarely present choices as flat, unstructured arrays. Instead, they leverage visual hierarchies, semantic categorizations, and physical spacing.¹⁴ These visual groupings allow users to rapidly bypass irrelevant options, invalidating the flat logarithmic selection assumption of the Hick-Hyman model.¹⁴
- **Visual Search Over Cognitive Choice:** Interface selection tasks are often dominated by visual search speeds and recognition processes rather than raw cognitive decision-making.¹⁵ Consequently, ocular movement models and visual foraging theories offer superior predictive power compared to the choice-entropy formulas of Hick-Hyman.¹⁵
- **Regression Variability:** Calibrating the empirical constant (b) in Hick-Hyman is highly context-dependent, showing high sensitivity to reading ability, interface familiarity, and semantic clarity, which limits its utility as a general analytical design metric.¹⁵

Empirical Research Methodologies and User Experience Metrics

Rigorous user experience research requires combining objective eye-tracking protocols with standardized subjective scales and structured heuristic evaluations.¹⁹

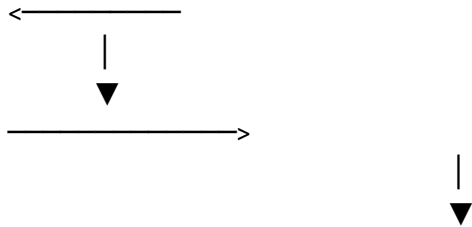
Eye-Tracking Experimental Procedures and Hardware Paradigms

Eye-tracking provides objective, real-time measures of visual attention, cognitive effort, and layout-driven friction, supplementing self-reported usability data.¹⁹



The gold standard for eye-tracking relies on invisible infrared light to detect pupil coordinates and track gaze direction via corneal reflections.²¹ However, emerging paradigms utilize high-frequency radar-based systems (operating between 60 GHz and 120 GHz) to capture eye movements.²¹ This radar approach operates in low-light conditions, functions through heavy eyewear, and does not require a direct line of sight, making it ideal for contextual testing.²¹ To evaluate design variants (such as in A/B testing), researchers employ a structured, sequential laboratory protocol ²¹:





Visual search efficiency is analyzed using specific metrics mapped to Areas of Interest (AOIs).²¹ Researchers apply mixed-effects statistical models to control for individual variation and isolate the exact cognitive costs of layout changes.²¹

- **Time to First Fixation (TTFF):** The elapsed time before a user's gaze first enters a target AOI.²¹ Longer TTFF values indicate poor visual hierarchy or weak layout-driven cues.²¹
- **Fixation Duration and Count:** The total time and number of stable gaze events within an AOI.²¹ Elevated values represent high visual friction, layout ambiguity, or difficulty processing visual information.²¹
- **Time to First Click (TTFC):** The elapsed time before a user registers their first physical click on a target.²¹ Discrepancies between TTFF and TTFC highlight action-pathway friction or lack of button signifiers.²¹
- **Saccadic Amplitude and Velocity:** Rapid eye movements between fixations.²¹ Higher peak velocities correlate with elevated emotional arousal, while scattered scanpaths represent disorienting layouts.²¹

Cognitive Workload and Subjective Usability Scales

Subjective metrics provide critical context to physiological eye-tracking data.¹⁹ The System Usability Scale (SUS), developed by John Brooke at Digital Equipment Corporation in the 1980s, remains a highly reliable post-test instrument.²² Comprising 10 Likert-scale questions, it yields a single usability score from 0 to 100.²² Crucially, a SUS score of 68 represents the global empirical average across over 500 studies compiled by Jeff Sauro, helping teams evaluate their system against industry benchmarks.²²

For complex, mission-critical workflows (such as healthcare, aerospace, or industrial controls), standard task-ease questions like the Single Ease Question (SEQ) are replaced by the multidimensional NASA Task Load Index (NASA-TLX).²² Administered immediately after a task, the NASA-TLX measures perceived workload across six key dimensions.²²

NASA-TLX DIMENSIONS	

- | | |
|-------------------|--|
| • Mental Demand | : How much mental and perceptual activity was required? |
| • Physical Demand | : How much physical activity was required? |
| • Temporal Demand | : How much time pressure did you feel? |
| • Performance | : How successful were you in accomplishing your goals? |
| • Effort | : How hard did you have to work to maintain performance? |
| • Frustration | : How insecure, discouraged, or stressed did you feel? |

Users rate each dimension on an unlabeled 21-point scale (ranging from Very Low to Very High) and complete pairwise comparisons to weight each dimension's relative contribution to overall task load.²² Because these subjective ratings can be influenced by individual bias or fatigue, they are most effective when paired with objective performance metrics, such as error rates and task completion times.²²

Scientific Method, Experimental Control, and Academic Writing Standards

Publishing high-quality HCI research requires adhering to systematic review frameworks, rigorous experimental design, and standardized manuscript formatting.²⁷

Hypothesis Generation and Experimental Control

To establish a solid theoretical foundation, researchers should utilize the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) framework to identify prior work and outline research trends.²⁸ Research questions must be evaluated against the FINER (Feasible, Interesting, Novel, Ethical, Relevant) criteria, while leveraging the PICO (Population, Intervention, Comparison, Outcome) format to define study variables.²⁹

PICO Framework = {Population, Intervention, Comparison, Outcome}

Experimental design requires mapping independent variables (the system variants) to dependent variables (the measured behaviors).³⁰ Confounds must be systematically controlled to preserve internal validity³⁰:

- **Visual Consistency:** Interface variants under test must maintain identical screen real estate, typographic hierarchies, and button styling.³⁰
- **System Capabilities:** Supported functions—such as search algorithms or keyboard shortcuts—must remain identical across all layouts.³⁰
- **Environmental Parity:** All participants must use equivalent hardware, input peripherals, and complete evaluations in identical lighting and environmental conditions.³⁰
- **Training and Experience:** Prior experience with the target system or visual domain must be balanced across participants to ensure comparable baseline skills.³⁰

Researchers must avoid "straw man comparisons"—where a new system is compared against an outdated or intentionally crippled interface.³⁰ Rigorous evaluations require comparing new designs against strong, state-of-the-art baselines representing standard industry conventions, demonstrating a clear, incremental contribution to the field.³⁰

Archival Academic Writing Formats and Standards

HCI research papers must follow structured layout templates depending on the target publisher or conference proceedings.²⁷

Structural Element	Taylor & Francis / General HCI Journal Template	ACM SIGCHI Proceedings Template
Header Configuration	Requires a shortened running head at the top of each page. ²⁷	Includes standard conference metadata, copyright blocks, and CCS concepts. ³³
Abstract Layout	Appears on page 1, identifying the work as theoretical, empirical, or methodological. ²⁷	Positioned directly below author metadata, summarizing key contributions and findings. ³²
Table of Contents	Required at the top of page 2, outlining the top three heading levels. ²⁷	Not required; papers use a continuous, multi-column layout. ³²
Heading Levels	Structured in an indented outline format (typically three levels maximum). ²⁷	Formatted using the acmart document class parameters. ³²
Footnotes & Notes	Compiled in a dedicated "NOTES" section following the main body text. ²⁷	Placed at the bottom of the page where referenced, managed via standard LaTeX syntax. ²⁷
Figure Captions	Listed sequentially on a separate page; captions must be self-explanatory and sentence-case. ²⁷	Placed directly below each figure; figures must include alt-text for accessibility. ²⁷

utilizing the "Libertine" font set.³³ To support the TAPS (The ACM Publication System) automated formatting workflow, authors must configure their LaTeX source files for each phase of the peer-review cycle.³²

```
Code snippet
% Initial Submission Format (Single-column, review line numbers, redacted authors)
\documentclass[manuscript,review,anonymous]{acmart}

% Camera-Ready Production Format (Double-column, final publication layout)
\documentclass[sigconf]{acmart}
```

When setting up Overleaf templates for initial review, authors should delete unnecessary files to ensure smooth compilation ³²:

- Delete unused engine-specific templates: sample-lualatex.tex, sample-sigconf.tex, sample-xelatex.tex.³²
- Delete local documentation files: README.txt, acmart.pdf.³²
- Work directly in sample-authordraft.tex (renaming it as appropriate) and link the custom .bib file.³²

During the review phase, papers must be formatted in a single-column layout using the manuscript option to facilitate annotation.³² Upon receiving conditional acceptance, authors transition the document to the double-column sigconf format.³² This transition changes the layout, requiring authors to manually adjust wide tables, equations, and figures to prevent column clipping.³² Submitted source files are then processed by TAPS to compile a high-resolution, double-column PDF alongside a responsive, accessible HTML5 version for the ACM Digital Library.³²

Collaborative Reference Management: Classic vs. New Workflows

Managing bibliographies in LaTeX can be streamlined using reference managers like Zotero, Mendeley, or Papers.³⁶ In March 2025, Overleaf released major citation management updates, introducing a more flexible, collaborative workflow.³⁷

Workflow Attribute	Classic Bulk Integration	New Inline Reference Search
Integration Mechanics	Imports the full personal or group library as a static, read-only .bib file. ³⁶	Dynamically pulls only cited references into a project-specific, editable .bib file. ³⁷

Editing Control	Only the user who linked the Zotero account can refresh or modify the bibliography. ³⁶	All project collaborators can search, add, and edit bibliography entries. ³⁷
Modification Autonomy	Formatting errors or citation keys must be resolved in the external reference manager. ³⁶	Citation keys and LaTeX formatting issues can be adjusted directly within Overleaf. ³⁷
Synch Reliability	Relies on manual full-library refreshes; prone to key mismatches if files are replaced. ³⁶	Automatic, incremental updates as references are added inline; prevents compilation errors. ³⁷

Mendeley Technical Integrations and Limitations

For researchers using Mendeley Desktop, syncing is enabled by checking "Enable BibTeX syncing" under preferences, which generates a dedicated .bib file for each folder collection.³⁸ However, this legacy workflow presents several technical challenges:

- **Forced Brace Capitalization:** Mendeley automatically wraps title fields in double braces ({{Title}}).³⁸ This forces literal capitalization and prevents BibTeX from applying standard sentence-case formatting, requiring manual post-export corrections.³⁸
- **Lack of "Unpublished" Document Types:** Mendeley lacks a native "Unpublished" document type, which is critical for researchers citing preprints from repositories like arXiv.org.³⁸ Authors must manually edit the exported .bib file to change document categories from "Generic" to "Unpublished" to prevent citation compilation failures.³⁸

Direct Web API and Custom Proxy Workflows

In multi-author academic collaborations, researchers can bypass local reference-manager limits by calling the Zotero Web API directly to fetch dynamically updated BibTeX files.³⁹ A standardized API call retrieves reference collections using specific IDs and access keys³⁹:

API URL = `[https://api.zotero.org/groups/GGGGGG/items/top?format=bibtex](https://api.zotero.org/groups/GGGGGG/items/top?format=bibtex)&key=KKKKKK`

Where GGGGGG is the Group ID and KKKKKK is the API access key.³⁹ To handle bibliographies that exceed the Zotero API's 100-item return limit, authors must append pagination parameters to retrieve subsequent references³⁹:

API URL (Paginated) = `[https://api.zotero.org/groups/GGGGGG/items/top?format=bibtex](https://api.zotero.org/groups/GGGGGG/items/top?format=bibtex)&key=KKKKKK&limit=100&start=100`

Alternatively, research teams can deploy a custom proxy server to coordinate API requests, manage pagination automatically, and convert Web API formats into Better BibTeX-compliant citation keys.³⁹ This proxy acts as an intermediary, providing a single, stable URL that compiles

and serves the complete bibliography directly to the Overleaf workspace.³⁹

Works cited

1. The most popular experience design trends of 2026 | by Joe Smiley ..., accessed on May 22, 2026, <https://uxdesign.cc/the-most-popular-experience-design-trends-of-2026-3ca85c8a3e3d>
2. AI-Driven Trends in UI/UX Design (2025–2026) - Medium, accessed on May 22, 2026, <https://medium.com/@designstudiouiux/ai-driven-trends-in-ui-ux-design-2025-2026-7cb03e5e5324>
3. Selecting a Subcommittee - ACM CHI 2026, accessed on May 22, 2026, <https://chi2026.acm.org/authors/papers/selecting-a-subcommittee/>
4. The Top 12 Annual UX Conferences in the US & EU - UXfolio Blog, accessed on May 22, 2026, <https://blog.uxfol.io/ux-conferences/>
5. UX Conference Guide for 2026 - Condens, accessed on May 22, 2026, <https://condens.io/guides/ux-conference-guide-2026/>
6. The 11 Must-Attend UX Conferences in 2025 - Dovetail, accessed on May 22, 2026, <https://dovetail.com/ux/best-ux-conferences/>
7. Top UI UX Design Conferences in 2026 - DesignRush, accessed on May 22, 2026, <https://www.designrush.com/agency/ui-ux-design/trends/ui-ux-conferences>
8. WRITING A SUCCESSFUL CHI PAPER (WORKSHOP IN THREE PARTS), accessed on May 22, 2026, <https://faculty.eng.ufl.edu/human-centered-computing/wp-content/uploads/sites/126/2019/11/hccatuf-chi-writing-2017-part03.pdf>
9. UX conferences that will satisfy your design wanderlust - Flying Bisons, accessed on May 22, 2026, <https://flyingbisons.com/blog/ux-conferences-that-will-satisfy-your-design-wanderlust>
10. Accepted Workshops - ACM CHI 2026, accessed on May 22, 2026, <https://chi2026.acm.org/workshops/accepted/>
11. How I Developed the 10 Usability Heuristics - UX Tigers, accessed on May 22, 2026, <https://www.uxtigers.com/post/usability-heuristics-history>
12. UX Laws | uxtoast, accessed on May 22, 2026, <https://www.uxtoast.com/ux-laws/>
13. Five HCI Laws for User Experience Design - MeasuringU, accessed on May 22, 2026, <https://measuringu.com/hci-laws/>
14. Hick's Law - Laws of UX, accessed on May 22, 2026, <https://lawsofux.com/hicks-law/>
15. Information Theoretic Models of HCI: A Comparison of the Hick-Hyman Law and Fitts' Law, accessed on May 22, 2026, https://www.researchgate.net/publication/228351073_Information_Theoretic_Models_of_HCI_A_Comparison_of_the_Hick-Hyman_Law_and_Fitts'_Law
16. Nielsen's Heuristics - The Decision Lab, accessed on May 22, 2026, <https://thedecisionlab.com/reference-guide/design/nielsens-heuristics>

17. 10 Usability Heuristics for UX Design - UX Pilot, accessed on May 22, 2026, <https://uxpilot.ai/blogs/usability-heuristics>
18. 10 Usability Heuristics for User Interface Design - NN/G, accessed on May 22, 2026, <https://www.nngroup.com/articles/ten-usability-heuristics/>
19. Eye tracking insights in UX research Q2 2025 - Tobii, accessed on May 22, 2026, <https://www.tobii.com/resource-center/scientific-publications/ux-research-2025-q2>
20. Fitts' Law as a Research and Design Tool in Human-Computer Interaction - York University, accessed on May 22, 2026, <https://www.yorku.ca/mack/hci1992.html>
21. Utilization of Eye-Tracking Metrics to Evaluate User Experiences—Technology Description and Preliminary Study - PMC, accessed on May 22, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC12526729/>
22. Beyond the NPS: Measuring Perceived Usability with the SUS, NASA-TLX, and the Single Ease Question After Tasks and Usability Tests - NN/G, accessed on May 22, 2026, <https://www.nngroup.com/articles/measuring-perceived-usability/>
23. Understanding Tobii Pro Lab's eye tracking metrics, accessed on May 22, 2026, <https://connect.tobii.com/s/article/understanding-tobii-pro-lab-eye-tracking-metrics>
24. Eye movements and user emotional experience: a study in interface design - PMC, accessed on May 22, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC11961869/>
25. Eye Tracking Metrics - WorldViz VR, accessed on May 22, 2026, <https://help.worldviz.com/sightlab/eye-tracking-metrics/>
26. Understanding the NASA-TLX: How Measuring Task Load Can Improve UX Design, accessed on May 22, 2026, <https://medium.com/@fs.sheikhahmadi/understanding-the-nasa-tlx-how-measuring-task-load-can-improve-ux-design-94bb2ad66b88>
27. HCI Style Guidelines for Final Submissions, accessed on May 22, 2026, <https://files.taylorandfrancis.com/hhciguideguidelines.pdf>
28. A Guide to Publishing Human-Computer Interaction (HCI) and Design Research Papers, accessed on May 22, 2026, <https://uxmag.com/articles/a-guide-to-publishing-human-computer-interaction-hci-and-design-research-papers>
29. Formulating the Research Question and Framing the Hypothesis - PMC - NIH, accessed on May 22, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC10353175/>
30. Experiments | Introduction to Human-Computer Interaction - Oxford Academic, accessed on May 22, 2026, <https://academic.oup.com/book/60808/chapter/529004175>
31. accessed on May 22, 2026, <https://lennartnacke.substack.com/p/how-to-write-a-chi-paper#:~:text=CHI's%20Ogates%20of%20steel%20and%20stone&text=The%20simple%20secret%20lies%20in,perspective%2C%20and%20deliver%20tangible%20contributions.>
32. CHI Publication Formats – CHI 2025, accessed on May 22, 2026, <https://chi2025.acm.org/chi-publication-formats/>
33. LaTeX templates and examples — Association for Computing Machinery (ACM) -

- Official Primary Article Templates - Overleaf, accessed on May 22, 2026,
<https://www.overleaf.com/gallery/tagged/acm-official>
34. Submissions - ACM, accessed on May 22, 2026,
<https://www.acm.org/publications/authors/submissions>
35. How to structure reports on experiments in human-computer interaction -
University of Calgary, accessed on May 22, 2026,
https://cspages.ucalgary.ca/~saul/hci_topics/assignments/controlled_expt/ass1_reports.html
36. Zotero | Overleaf docs, accessed on May 22, 2026,
<https://docs.overleaf.com/integrations-and-add-ons/reference-manager-integrations/zotero>
37. Managing citations and bibliographies in Overleaf just got easier, accessed on
May 22, 2026,
<https://www.overleaf.com/blog/managing-citations-and-bibliographies-in-overleaf-just-got-easier>
38. HOWTO: Use Mendeley to create citations using LaTeX and BibTeX., accessed on
May 22, 2026,
<https://blog.mendeley.com/2011/10/25/howto-use-mendeley-to-create-citations-using-latex-and-bibtex/>
39. Automated Bibliography Management in Overleaf with Zotero and a Custom
Proxy, accessed on May 22, 2026,
<https://upb-syssec.github.io/blog/2025/zotero-overleaf-integration/>